1. а) $A=\{1;\frac{1}{2};\frac{1}{4};\frac{1}{8};\dots\}=\{\frac{1}{2^n} \mid n\in[0;+\infty)_{n\in\mathbb{Z}}\}$

б) $A=\{\frac{2}{3};\frac{5}{9};\frac{26}{27};\frac{80}{81};\dots\}=\{\frac{3^n-1}{3^{n+1}} \mid n\in[0;+\infty)_{n\in\mathbb{Z}}\}$

в) $A=\{\frac{1}{2};-\frac{1}{3};\frac{1}{4};-\frac{1}{5};\dots\}=\{(-1)^n\frac{1}{n+2} \mid n\in[0;+\infty)_{n\in\mathbb{Z}}\}$

г) $A=\{0;\frac{1}{3};\frac{1}{2}=\frac{2}{4};\frac{3}{5};\dots\}=\{\frac{n}{n+2} \mid n\in[0;+\infty)_{n\in\mathbb{Z}}\}$

2. $A=\{-1\le x^2+31\le 256\}=\{-32\le x^2\le 225\}=\{x \mid x\in[-15;15]\}$

$B=\{x^3-10x^2+16x\le 0\}=\{x(x-8)(x-2)\le 0\}=\{x \mid x\in(-\infty;0]\cup[2;8]\}$

$C=\{x^3-10x^2+16x=0\}=\{0;2;8\}$

$B\cup C=B=\{x \mid x\in(-\infty;0]\cup[2;8]\}$

$(B\cap C)\triangle A=\{x \mid x\in[-15;0)\cup(0;2)\cup(2;8)\cup(8;15]\}$

$\overline{A\setminus C}=\{x \mid x\in(-\infty;-15)\cup\{0;2;8\}\cup(15;+\infty)\}$

3. $U=\{-5;-4;\dots;5\}$

$A=\{x: f(x)=0 \text{ и } x\in\mathbb{Z}\}$ $\quad \overline{A}-?$ $\quad |A|-?$ $\quad A\triangle U-?$ $\quad |P(A)|-?$

а) $f(x)=x^4+x^3-6x^2=x^2(x-2)(x+3)\Rightarrow$

$\Rightarrow A=\{-3;0;2\}$

$\overline{A}=\{-5;-4;-2;-1;1;3;4;5\}$

$A\triangle U=\{-5;-4;-2;-1;1;3;4;5\}$

$|A|=3 \quad |P(A)|=8\ (2^{|A|})$

б) $f(x)=x^3-6x^2+11x-6$

$f(+2)=8-24+22-6=0$

$f(1)=1+11-6-6=0$

$f(3)=27-54+33-6=0$

$A=\{1;2;3\}$

$\overline{A}=\{-5;-4;-3;-2;-1;0;4;5\}$

$A\triangle U=\overline{A}$; $|A|=3$; $|P(A)|=8$

б) $f(x) = x^4 - x^3 - 6x^2 - x + 3$

$f(-1) = 1 + 1 - 6 + 1 + 3 = 0$

$f(3) = 81 - 27 - 54 - 3 + 3 = 0$

$x^4 - x^3 - 6x^2 - x + 3 \;\big|\; x^2 - 2x - 3$

$x^4 - 2x^3 - 3x^2 \;\big|\; x^2 + x - 1$

$x^3 - 3x^2 - x$

$x^3 - 2x^2 - 3x$

$-x^2 + 2x + 3$

$-x^2 + 0x - 3$

0

$x = \frac{-1 \pm \sqrt{5}}{2} \notin \mathbb{Z}$

$A = \{-1; 3\}$

$\bar{A} = \{-5; -4; -3; -2; 0; 1; 2; 4; 5\}$

$A \triangle U = \bar{A}$

$|A| = 2 \quad |P(A)| = 4$

г) $f(x) = x^3 - 5x^2 - 2x + 24 = (x+2)(x-3)(x-4)$

	1	-5	-2	24
2	1	-3	-8	8
-2	1	-7	12	0
3	1	-4	0	
4	1	0		

$A = \{-2; 3; 4\}$

$A \triangle U = \bar{A} = \{-5; -4; -3; -1; 0; 1; 2; 5\}$

$|A| = 3 \quad |P(A)| = 8$

4. $|P(A)| = 32 = 2^{|A|} \Rightarrow |A| = 5$

$|P(B)| = 16 = 2^{|B|} \Rightarrow |B| = 4$

$|P(C)| = 1024 = 2^{|C|} \Rightarrow |C| = 10$

$|A \times B \times C| = |A| \cdot |B| \cdot |C| = 200$

5. $|A \times B| - ?$

$A = \{1; 2; \dots; 10\}$

$B = \{1; 3; 5\}$

$\Rightarrow |A \times B| = |A| \cdot |B| = 30$